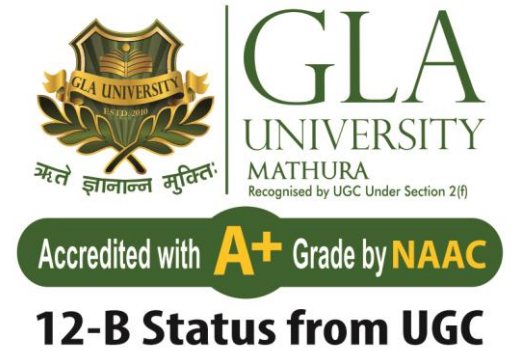


3rd PPT

Timing Diagram



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Outline

- Machine Cycle, T- State and Instruction Cycle
- Timing Diagram for Different Machine Cycle
- Timing diagram for the Instructions: MVI B, 43 and STA 526A

Machine Cycle, T State and Instruction Cycle

Timing Diagram is a graphical representation. Timing diagram of 8085 instructions represents the execution time taken by each instruction in a graphical format. The execution time is represented in T-states.

T-State:

- The processor takes a definite time to execute the machine cycles. The time taken by the processor to execute a machine cycle is expressed in T-states.
- A portion of an operation carried out in one system clock period is called as T-state.
- One T-state is equal to the time period of the internal clock signal of the processor. The T-state starts at the falling edge of a clock

Machine Cycles Of 8085 Microprocessor

The time required to access the memory or input/output devices is called machine cycle.

Machine cycles of 8085:

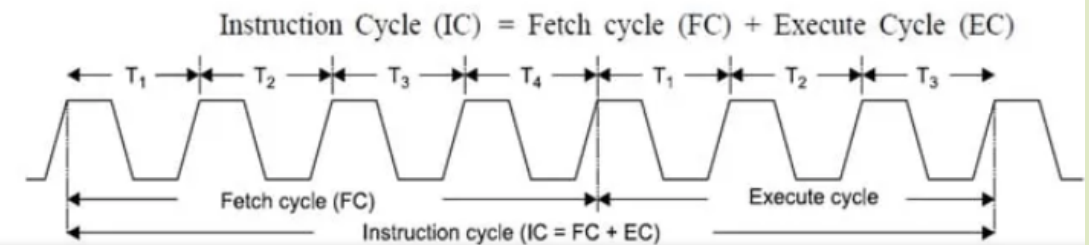
The 8085 microprocessor has 5 basic machine cycles. They are

1. Opcode fetch cycle (4T)
2. Memory read cycle (3 T)
3. Memory write cycle (3 T)
4. I/O read cycle (3 T)
5. I/O write cycle (3 T)

Each instruction of the 8085 processor consists of one to five machine cycles, i.e., when the 8085 processor executes an instruction, it will execute some of the machine cycles in a specific order.

Instruction Cycle

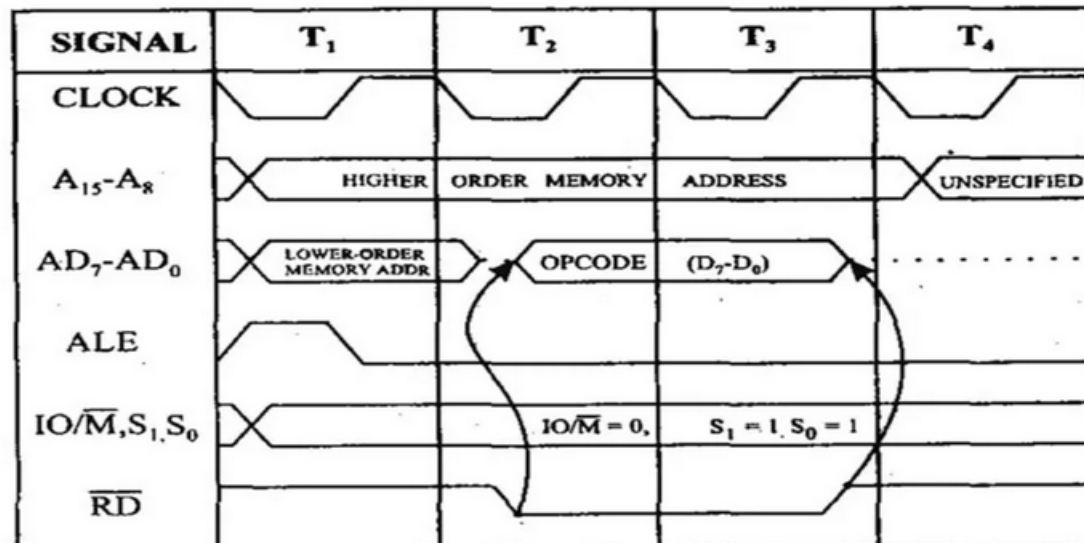
The time required to execute an instruction is called instruction cycle.



Machine Cycles of 8085

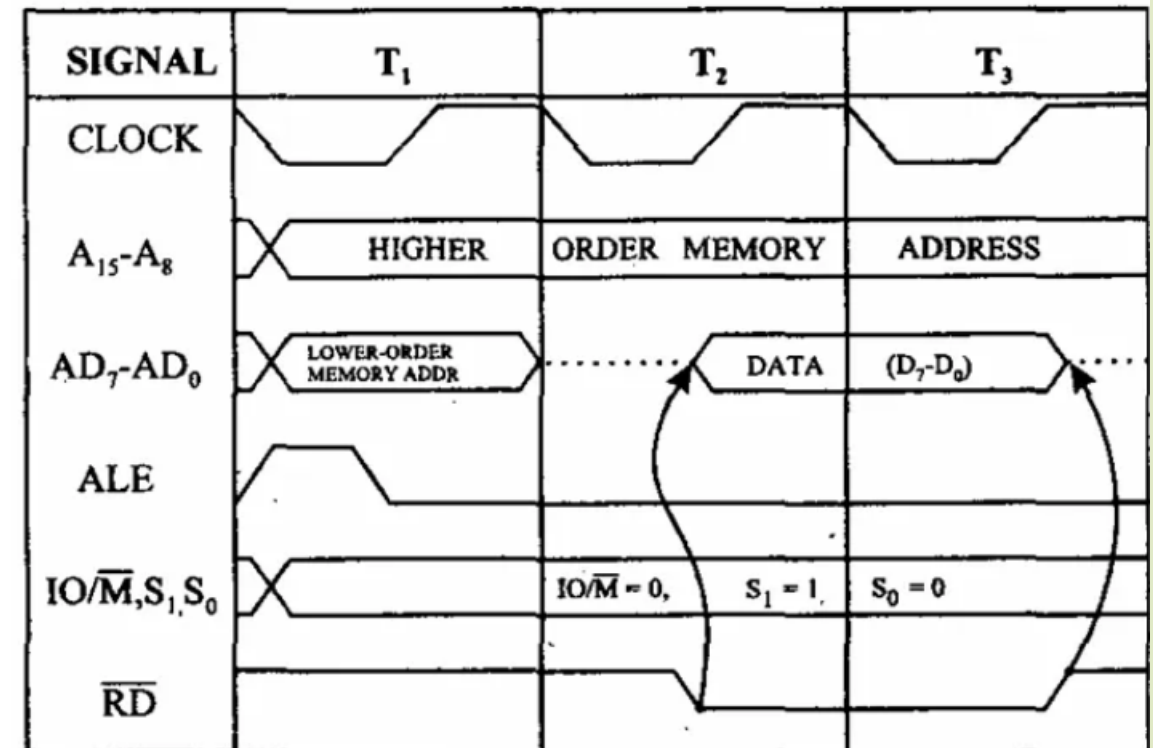
Opcode Fetch Machine Cycle Of 8085

- Each instruction of the processor has one byte opcode.
- The opcodes are stored in memory. So, the processor executes the opcode fetch machine cycle to fetch the opcode from memory.
- Hence, every instruction starts with opcode fetch machine cycle.
- The time taken by the processor to execute the opcode fetch cycle is 4T.
- In this time, the first, 3 T-states are used for fetching the opcode from memory and the remaining T-states are used for internal operations by the processor.



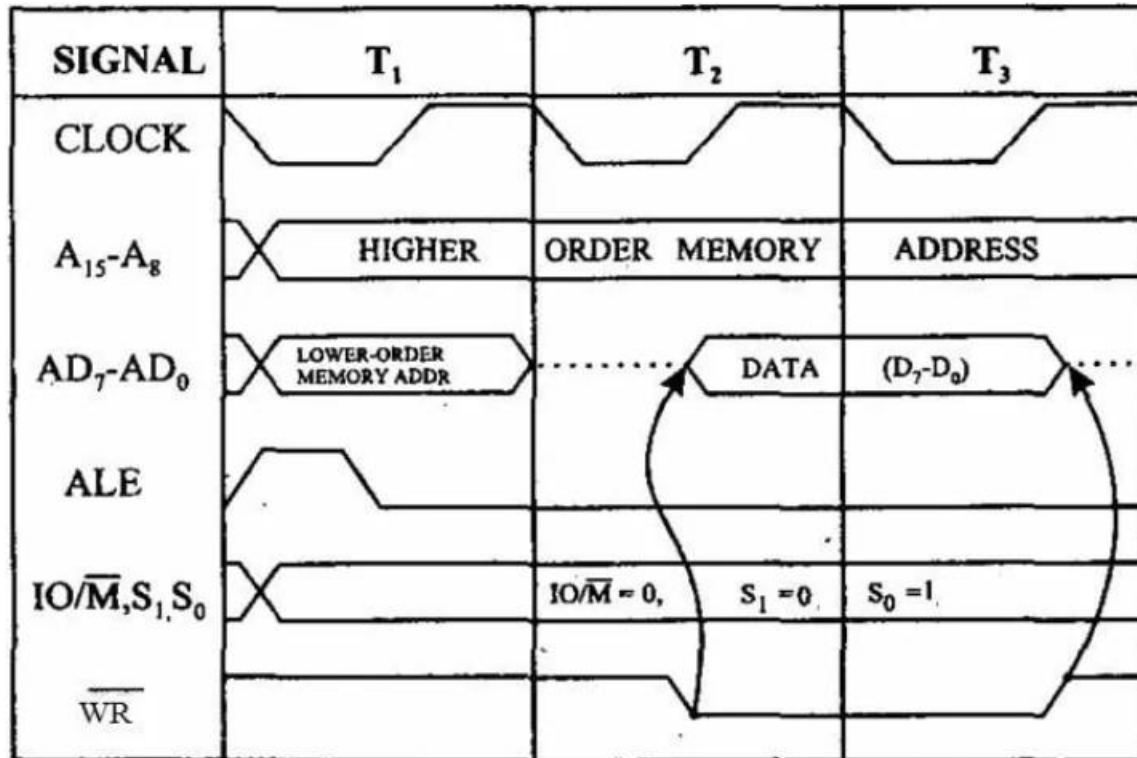
Memory Read Machine Cycle Of 8085

- The memory read machine cycle is executed by the processor to read a data byte from memory. The processor takes 3T states to execute this cycle.
- The instructions which have more than one byte word size will use the machine cycle after the opcode fetch machine cycle.



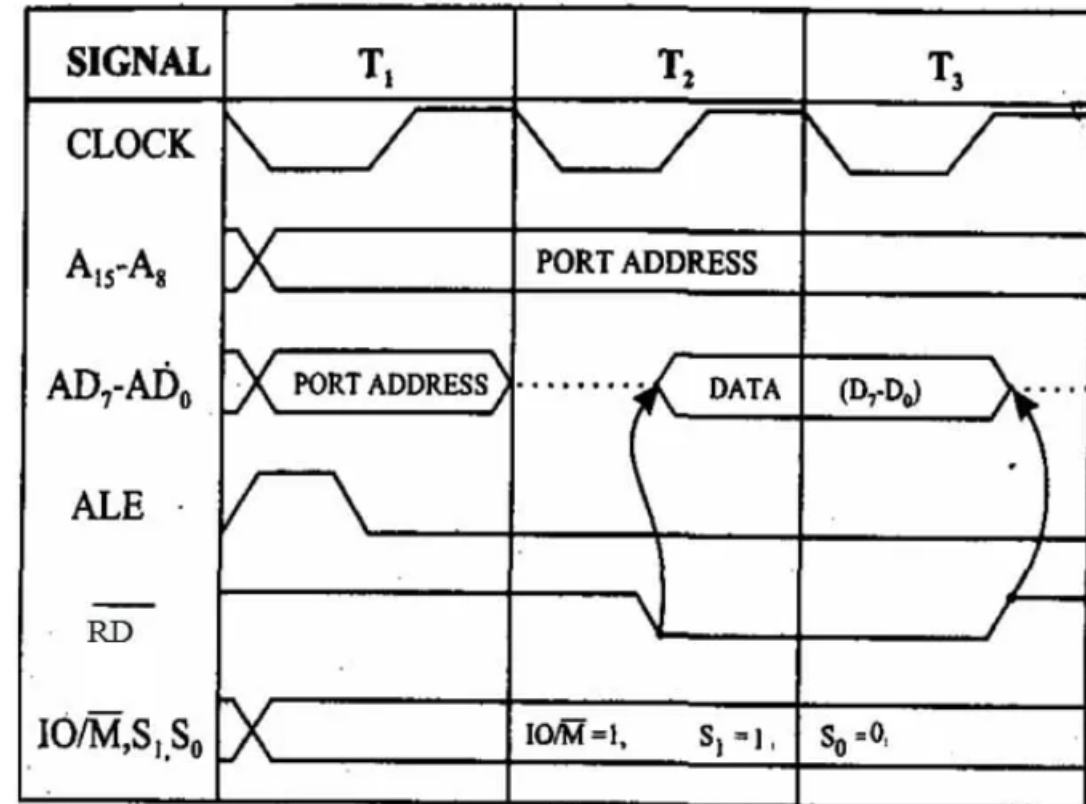
Memory Write Machine Cycle Of 8085

- The memory write machine cycle is executed by the processor to write a data byte in memory. The processor takes 3T states to execute this cycle.
- The instructions which have more than one byte word size will use the machine cycle after the opcode fetch machine cycle.



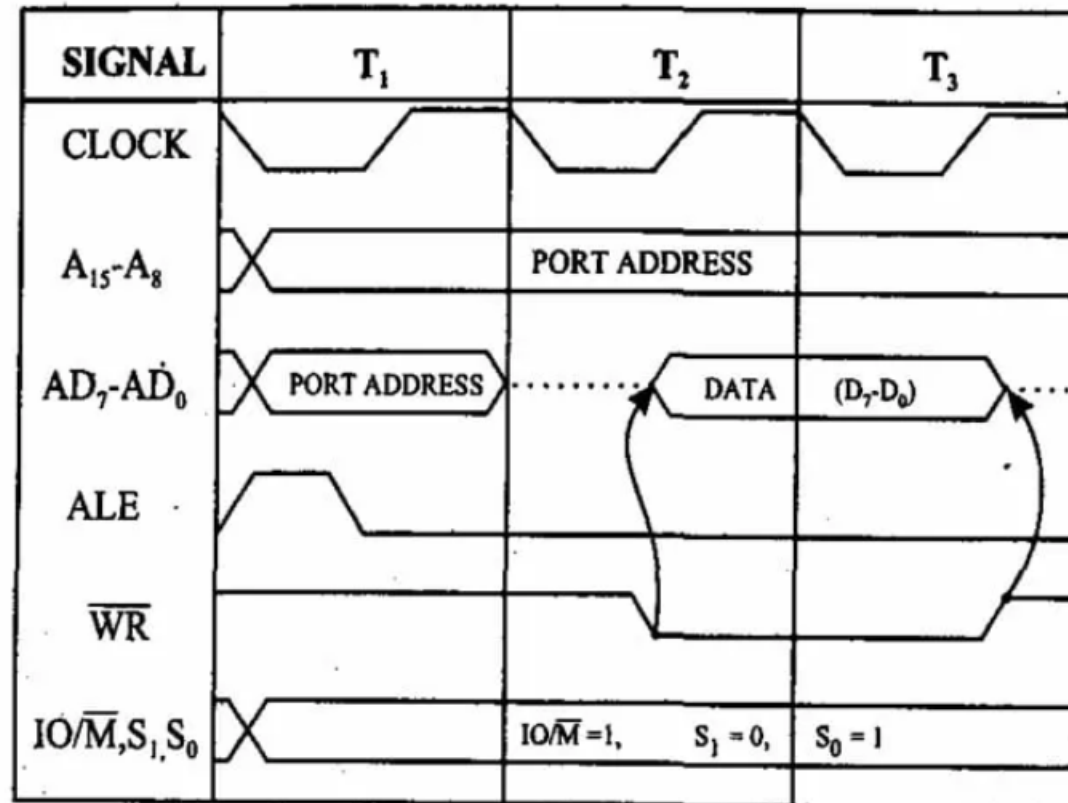
I/O Read Machine Cycle Of 8085

- The I/O Read machine cycle is executed by the processor to Read a data byte from the I/O port or from a peripheral, which is I/O, mapped in the system.
- The processor takes 3T states to execute this machine cycle.



I/O Write Machine Cycle Of 8085

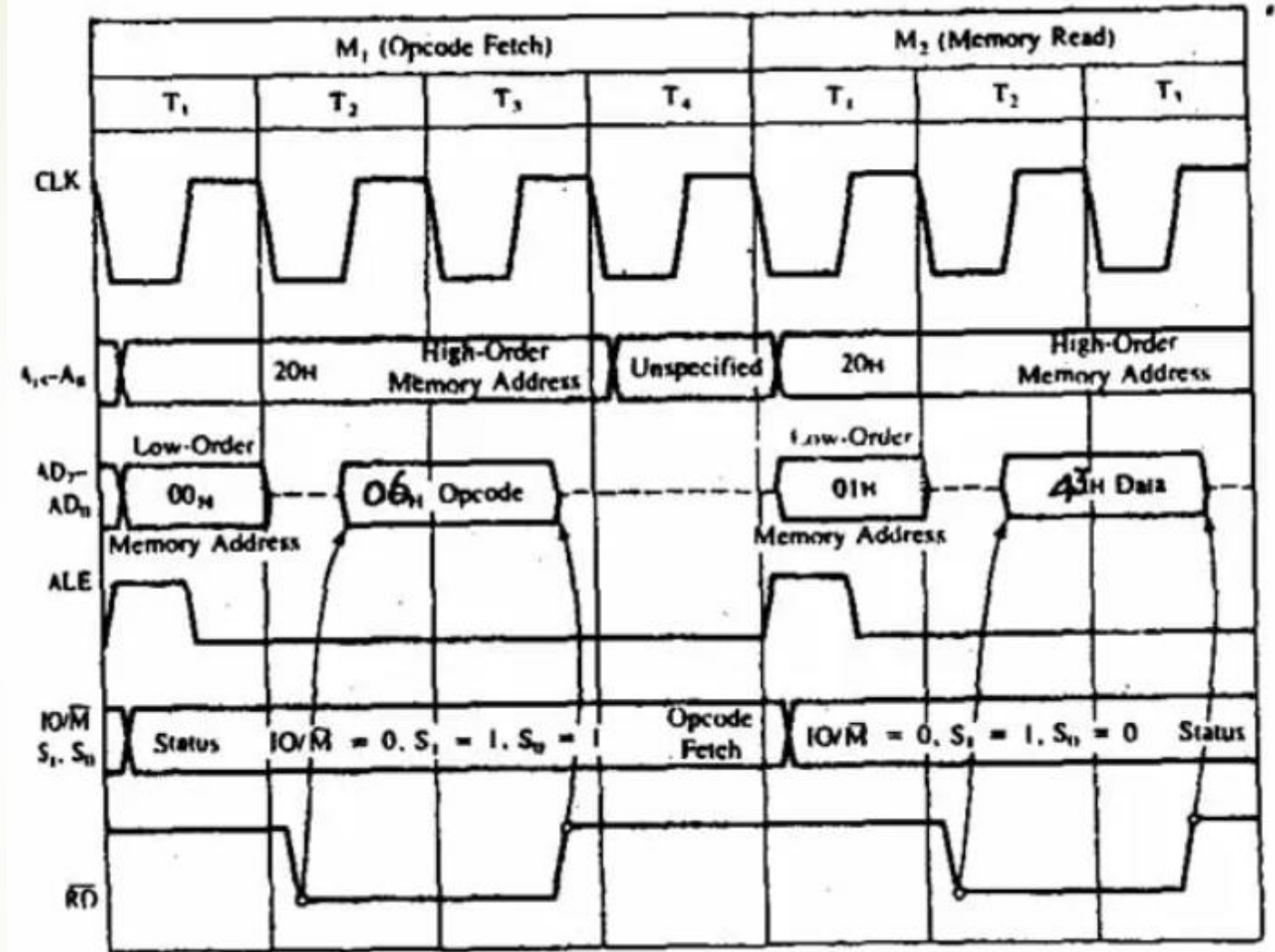
- The I/O write machine cycle is executed by the processor to write a data byte in the I/O port or to a peripheral, which is I/O, mapped in the system.
- The processor takes 3T states to execute this machine cycle.



Timing Diagram For MVI B, 43H

- Fetching the opcode 06H from the memory 2000H. (Opcode fetch machine cycle)
- Read (move) the data 43H from memory 2001H. (memory read).

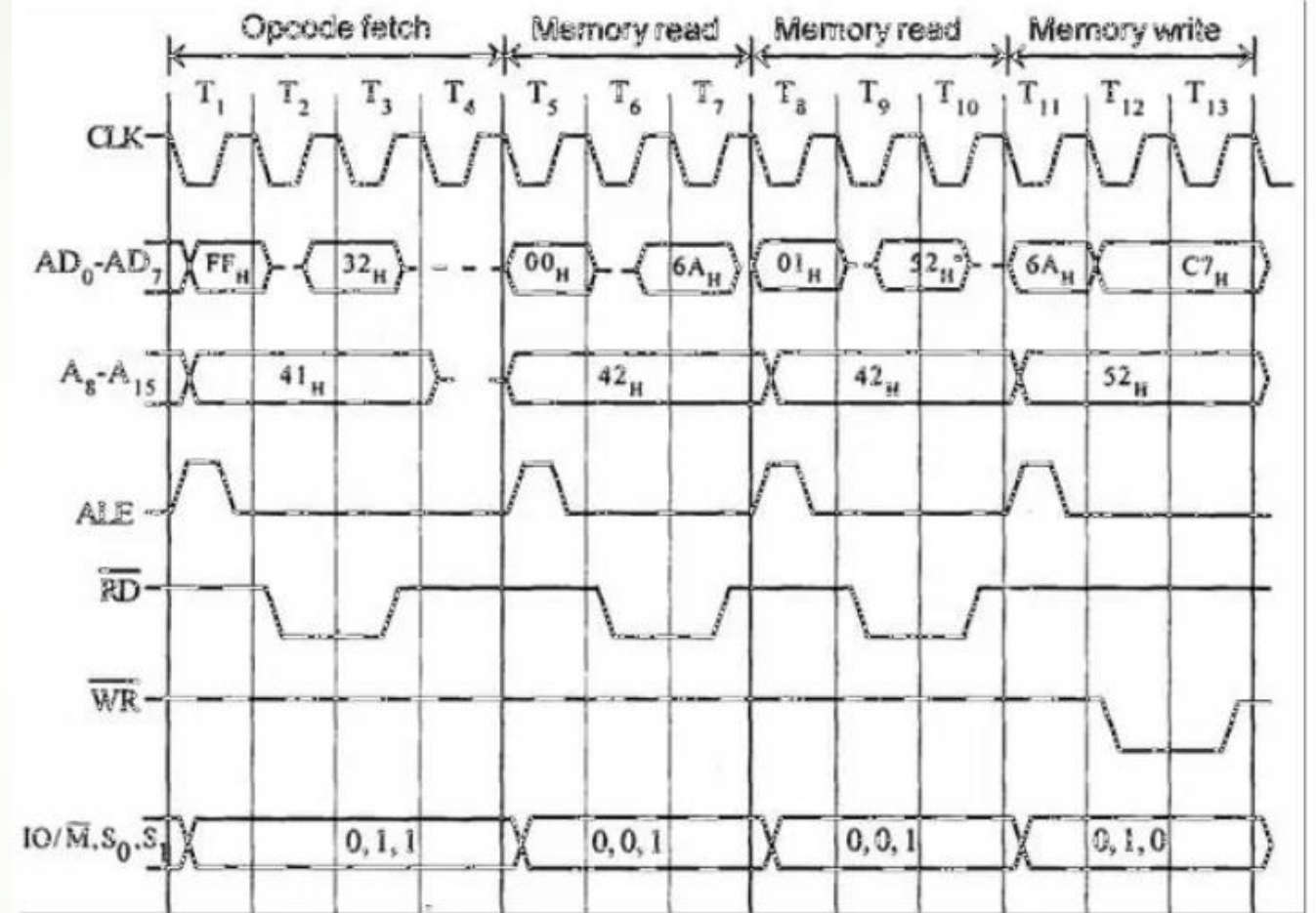
Address	Mnemonics	Opcode
2000H	MVI B,43H	06H
2001H		43H



Timing Diagram For STA 526AH

- STA means Store Accumulator -The contents of the accumulator are stored in the specified address (526A).
- The opcode of the STA instruction is said to be 32H. It is fetched from the memory 41FFH (see fig). – *OF machine cycle*.
- Then the lower order memory address is read(6A). – *Memory Read Machine Cycle*
- Read the higher order memory address (52) – *Memory Read Machine Cycle*.
- The combinations of both the addresses are considered and the content from accumulator is written in 526A. – *Memory Write Machine Cycle*
- Assume the memory address for the instruction and let the content of accumulator is C7H. So, C7H from accumulator is now stored in 526A.

Address	Mnemonics	Opcode
41FFH	STA 526A	32H
4200H		6AH
4201H		52H



Questions

Calculate the Instruction size, Machine Cycle, and T-States for the following instructions. Also write the name of Machine cycle.

1. MOV A,B
2. ADD B
3. LDA 2030
4. LXI H, 3425
5. CMP
6. SHLD 2B3A
7. LHLD 2000
8. IN 45
9. OUT 33
10. CALL 3041

Draw the Timing diagram of the following Instructions.

1. MVI C, 3B
2. LDA 3C40
3. LXI H, 3425
4. SHLD 20A0
5. LHLD 3031
6. IN 40
7. OUT 30